

Electrostatic Interactions in the Post-Ewald Era: Theoretical Insights and Machine Learning Treatments

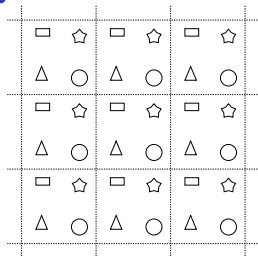
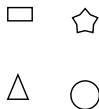
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March 11, 2026

Electrostatic interactions in periodic systems

Periodic Boundary Conditions(PBC)¹



Electrostatic Energy Calculation Under PBC:

$$E = \frac{1}{2} \sum_{i=1}^{\infty} \sum_{j=1, j \neq i}^{\infty} q_i q_j \frac{1}{r_{ij}} \quad \longrightarrow \quad E = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N q_i q_j \left(\sum'_{\mathbf{n} \in \mathbb{Z}^3} \frac{1}{|\mathbf{r}_{ij} + \mathbf{nL}|} \right)$$

The periodic extension of the Coulomb potential leads to a **conditionally convergent** series.

$$S_A = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \dots = \ln 2$$

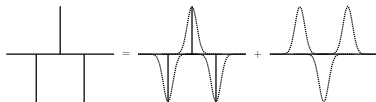
$$S_A = (1 - \frac{1}{2}) - \frac{1}{4} + (\frac{1}{3} - \frac{1}{6}) - \frac{1}{8} + \dots = \frac{1}{2} (1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \dots) = \frac{1}{2} \ln 2$$

¹M. P. Allen and D. J. Tildesley, *Computer Simulation of Liquids*, 2nd ed., Oxford University Press, 2017

Ewald Summation Method

P. P. Ewald splits the Coulomb potential²

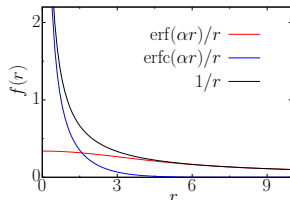
$$\frac{1}{r} = \frac{\operatorname{erfc}(\alpha r)}{r} + \frac{\operatorname{erf}(\alpha r)}{r}$$



Traditional Ewald energy:

$$E = \frac{1}{2} \sum_{\mathbf{n} \in \mathbb{Z}^3} \sum_i^N \sum_j^N q_i q_j \frac{\operatorname{erfc}(\alpha |\mathbf{r}_{ij} + \mathbf{nL}|)}{|\mathbf{r}_{ij} + \mathbf{nL}|} - \frac{\alpha}{\sqrt{\pi}} \sum_i^N q_i^2 \quad \text{R-space sum}$$

$$+ \frac{1}{2V} \sum_i^N \sum_j^N q_i q_j \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} e^{-\frac{k^2}{4\alpha^2}} e^{i\mathbf{k} \cdot \mathbf{r}_{ij}} \quad \text{k-space sum}$$



Remaining boundary term :

$$E^{IB} = \frac{1}{2V} \sum_i^N \sum_j^N q_i q_j \lim_{\mathbf{k} \rightarrow 0} \frac{4\pi}{k^2} e^{-\frac{k^2}{4\alpha^2}} e^{i\mathbf{k} \cdot \mathbf{r}_{ij}}$$

The Ewald method is the **gold standard** in electrostatic computation.

²P. P. Ewald, *Ann. Phys.* **64**, 253 (1921)

The Post Ewald Era

Pitfalls and Misconceptions of the Ewald Method:

- ▶ The physical picture of the k -space sum is not intuitive.
- ▶ Inherently tied to PBC, which can affect simulation outcomes.
- ▶ Low computational efficiency $\mathcal{O}(N^2)$.
- ▶ Contains an arbitrary parameter α .

Therefore, electrostatic energy calculations have diverged into two main directions:

- ▶ Abandoning Ewald's concept, developing purely real-space electrostatics.
- ▶ Building upon Ewald's concept, focusing on computational efficiency.

I . A Unified Theoretical Framework

Abandoning Ewald's concept and developing purely real-space electrostatics introduces the following issues:

- ▶ Can these new electrostatic interactions correctly reproduce the physical properties they govern?
- ▶ Can this correctness be determined directly from their functional forms?

Thus, a unified theoretical framework connecting **functional forms** and **physical properties** is necessary³.

³Y. Zhao and Z. Hu, *J. Chem. Theory Comput.* **21**, 5916 (2015)

I. The Interaction Benchmark and Its Fundamental Defect

The Ideal Benchmark

The complete electrostatic interaction is the fundamental benchmark for any physical property:

$$u_{\text{pair}}(\mathbf{r}) = \frac{1}{r} + \sum_{\mathbf{n} \neq 0} \left(\frac{1}{|\mathbf{r} + \mathbf{n}\mathbf{L}|} - \frac{1}{|\mathbf{n}\mathbf{L}|} \right)$$

Termwise Rearrangement:

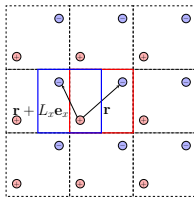
$$u_{\text{pair}}^{\text{red}}(\mathbf{r}) = u_{\text{pair}}(\mathbf{r}), \quad u_{\text{pair}}^{\text{blue}}(\mathbf{r}) = u_{\text{pair}}(\mathbf{r} - L_x \mathbf{e}_x)$$

Energy of an Identical Charge in an Identical Lattice:

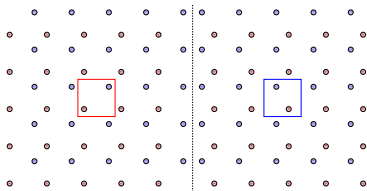
$$\Delta u \sim L^{-1} \neq 0$$

The benchmark is non-unique.

An Intrinsic Physical Defect



Reclassification of unit cells:



Therefore, electrostatic interactions have a rigorously correct form, but it is not directly usable in practical simulations.

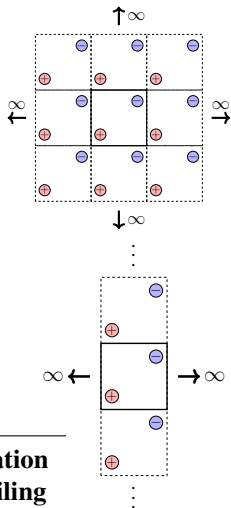
I. Traditional Ewald Interaction

$$u_{\text{pair}}^{tf}(\mathbf{r}) = \frac{\text{erfc}(\alpha r)}{r} + \sum_{\mathbf{n} \neq 0} \left(\frac{\text{erfc}(\alpha |\mathbf{r} + \mathbf{nL}|)}{|\mathbf{r} + \mathbf{nL}|} - \frac{\text{erfc}(\alpha |\mathbf{nL}|)}{|\mathbf{nL}|} \right) + \frac{2\alpha}{\sqrt{\pi}} + \frac{1}{V} \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} e^{-\frac{k^2}{4\alpha^2}} (e^{i\mathbf{k} \cdot \mathbf{r}} - 1)$$

$$u_{\text{pair}}^{ib}(\mathbf{r}) = -\frac{2\pi}{V} \lim_{\mathbf{k} \rightarrow 0} \frac{(\mathbf{k} \cdot \mathbf{r})^2}{|\mathbf{k}^2|} = \frac{1}{2V} \lim_{\Omega \rightarrow \infty} \oint_{\partial\Omega} dS \mathbf{n} \cdot \left[\mathbf{r} \cdot \left(\mathbf{r} \cdot \nabla_{\mathbf{a}} \frac{1}{|\mathbf{a}|} \right) \right]$$

$$= \begin{cases} \text{cube-shaped } \Omega & \xrightarrow{+u_{\text{pair}}^{tf}} u_{\text{pair}}^{ib, \text{cube}}(\mathbf{r}) = u_{\text{pair}}^{tf}(\mathbf{r}) - \frac{2\pi}{3V} r^2 \\ \text{plate-shaped } \Omega & \xrightarrow{+u_{\text{pair}}^{tf}} u_{\text{pair}}^{ib, \text{plate}}(\mathbf{r}) = u_{\text{pair}}^{tf}(\mathbf{r}) - \frac{2\pi}{V} z^2 \\ \text{other shape} & \xrightarrow{+u_{\text{pair}}^{tf}} u_{\text{pair}}^{ib, \text{shape}}(\mathbf{r}) = u_{\text{pair}}^{tf}(\mathbf{r}) + \dots \end{cases}$$

$$u_{\text{pair}}^{tf}(\mathbf{r}) : \text{ a rearrangement of } \frac{1}{r} + \sum_{\mathbf{n} \in \mathbb{Z}^3, \mathbf{n} \neq 0} \left(\frac{1}{|\mathbf{r} + \mathbf{nL}|} - \frac{1}{|\mathbf{nL}|} \right) - u_{\text{pair}}^{ib}(\mathbf{r})$$



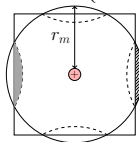
The Ewald interaction class results from a definitive summation order mathematically, and its form embodies the crystal's tiling rules (shape) physically.

I. Purely Real-Space Interactions without k-Space

Angular-Averaged Ewald interaction⁴

$$u_{\text{pre-aa}}(r) = \frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta u_{\text{pair}}^{\text{tf}}(\mathbf{r})$$

$$= \begin{cases} \frac{1}{r} + \frac{r^2}{2r_m^3} - \frac{3}{2r_m} & r \leq r_m \\ 0 & r > r_m \end{cases}$$



$$\frac{4\pi}{3} r_m^3 = V_{\text{cell}}$$

It's periodic extension constitutes:

$$u_{\text{pair}}^{\text{aa}}(\mathbf{r}) = u_{\text{pre-aa}}(r) + \sum_{\mathbf{n} \neq 0} u_{\text{pre-aa}}(|\mathbf{r} + \mathbf{nL}|)$$

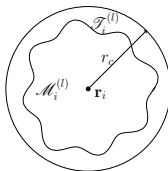
We provide a rigorous derivation in a single step, that is, starting from the infinite series form.

⁴E. Yakub and C. Ronchi, *J. Chem. Phys.* **119**, 11556 (2003)

⁵I. Fukuda, *J. Chem. Phys.* **139**, 174107 (2013)

Zero-Multipole interaction⁵

$$E_{\text{ZM}} = \frac{1}{2} \sum_{i=1}^N \sum_{j \in \mathcal{M}_i^{(l)}} q_i q_j u(\mathbf{r}_{ij})$$



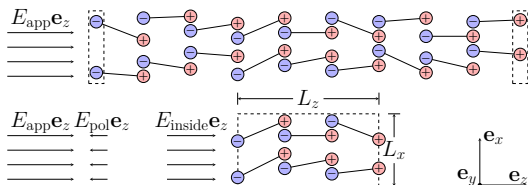
$\mathcal{M}_i^{(l)}$: the set of charge indices such that the moments up to the l -th order in its union with i are zero.

$$E_{\text{ZM}} \approx \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i, r_{ij} < r_c} q_i q_j u_{\text{pair}}^{\text{zm},l}(r_{ij})$$

$u_{\text{pair}}^{\text{zm},l} \equiv$ it's periodic extension

A derivation error exists after approximation, but not for orders $l \leq 2$ and below.

I. Dielectric Response and Boundary Term Form



Linear response :

$$\mathbf{M}_{\text{ind}}/V = \epsilon_0(\epsilon - 1)(\mathbf{E}_{\text{app}} - \mathbf{E}_{\text{pol}})$$

Depolarization field :

$$\mathbf{E}_{\text{pol}} = -\gamma \mathbf{M}_{\text{ind}}/V\epsilon_0$$

$$\gamma = \begin{cases} 1/3 & \text{cube} \Leftrightarrow \text{ib, cube} & u_{\text{pair}}^{\text{ib, cube}} \rightarrow \epsilon_0 \frac{\mathbf{M}_{\text{ind}}^{\text{ib, cube}}}{3V} = \frac{(\epsilon - 1)}{(\epsilon + 2)} \mathbf{E}_{\text{app}} \\ 1 & \text{plate} \Leftrightarrow \text{ib, plate} & u_{\text{pair}}^{\text{ib, plate}} \rightarrow \epsilon_0 \frac{\mathbf{M}_{\text{ind}}^{\text{ib, plate}}}{V} = \frac{(\epsilon - 1)}{\epsilon} \mathbf{E}_{\text{app}} \\ 0 & \text{no-boundary} \Leftrightarrow \text{tf} & u_{\text{pair}}^{\text{tf}} \rightarrow \frac{\mathbf{M}_{\text{ind}}^{\text{tf}}}{V} = \epsilon_0(\epsilon - 1) \mathbf{E}_{\text{app}} \end{cases}$$

Similarly for other shapes (γ).

model & spring constants	$E_{\text{app}} (\text{V}/\text{\AA})$	tf		ib, plate		ib, cube	
		$\frac{M_{\text{ind}}^{\text{tf}}}{V\epsilon_0} (\text{V}/\text{\AA})$	ϵ_{∞}	$\frac{M_{\text{ind}}^{\text{ib, plate}}}{V\epsilon_0} (\text{V}/\text{\AA})$	ϵ_{∞}	$\frac{M_{\text{ind}}^{\text{ib, cube}}}{V\epsilon_0} (\text{V}/\text{\AA})$	ϵ_{∞}
ion, κ	0.1	0.03654	1.3654	0.02751	1.3794	0.03369	1.3795
	0.2	0.07308		0.05501		0.06738	
mol, κ	0.1	0.03075	1.3075	0.02352	1.3075	0.02789	1.3075
	0.2	0.06150		0.04704		0.05579	

We began by establishing Geometric Shape is physically equivalent to Lattice Tiling Rule

I . Dielectric Response and Boundary Term Form

Under adiabatic :

$$H_{\mathbf{E}}(\bar{\mathbf{f}}, \mathbf{E}) = H_0(\bar{\mathbf{f}}, \bar{\mathbf{e}}_0(\bar{\mathbf{f}}, \mathbf{E})) - \mathbf{E} \cdot \mathbf{M}(\bar{\mathbf{f}}, \mathbf{E})$$

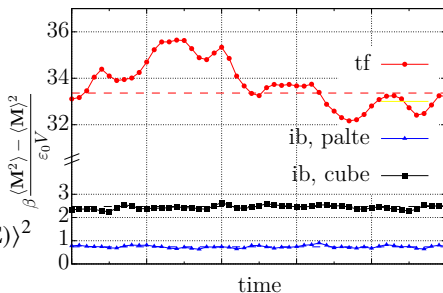
$$\mathbf{M}(\mathbf{E}) = \frac{1}{Q(\mathbf{E})} \int \mathbf{M}(\bar{\mathbf{f}}, \mathbf{E}) e^{-\beta H_{\mathbf{E}}(\bar{\mathbf{f}}, \mathbf{E})} d\bar{\mathbf{f}}$$

$\Downarrow$

$$\frac{\partial \mathbf{M}(\mathbf{E})}{\partial \mathbf{E}} = \langle \frac{\partial \mathbf{M}(\bar{\mathbf{f}}, \mathbf{E})}{\partial \mathbf{E}} \rangle + \beta \langle \mathbf{M}^2(\bar{\mathbf{f}}, \mathbf{E}) \rangle - \beta \langle \mathbf{M}(\bar{\mathbf{f}}, \mathbf{E}) \rangle^2$$

$$\partial \mathbf{M} = \mathbf{M}_{\text{ind}} \quad \Downarrow \quad \partial \mathbf{E} = \mathbf{E}_{\text{app}}$$

$$\beta \frac{\langle \mathbf{M}^2 \rangle - \langle \mathbf{M} \rangle^2}{\epsilon_0 V} = \begin{cases} \frac{\epsilon - 1}{\epsilon + 2} - \frac{\epsilon_\infty - 1}{\epsilon_\infty + 2}, & u_{\text{pair}}^{\text{ib, cube}} \\ \frac{\epsilon - 1}{\epsilon} - \frac{\epsilon_\infty - 1}{\epsilon_\infty}, & u_{\text{pair}}^{\text{ib, plate}} \\ \epsilon - \epsilon_\infty, & u_{\text{pair}}^{\text{tf}} \\ \dots, & u_{\text{pair}}^{\text{other-shape}} \end{cases}$$



- Unification of PBC dielectric response with macroscopic theory.
- Elimination of artifacts in the traditional reaction field (RF)⁶⁷.

In classical MD, $\epsilon_\infty = 1$.

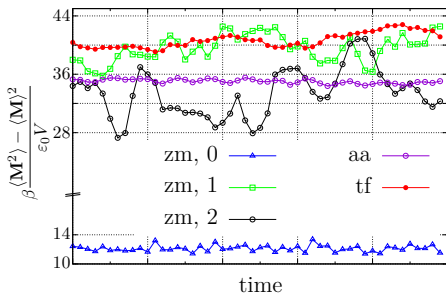
⁶S. W. de Leeuw and E. R. Smith, *Proc. R. Soc. Lond. A* **373**, 57 (1980)

⁷D. Frenkel and B. Smit, *Understanding Molecular Simulation*, Academic Press,

I. Dielectric Response and Boundary Term Form

- $zm, 0$: Non-zero dipole in cutoff $\rightarrow$ resembles IB (induces E_{pol}).
- $zm, 2$: Potential deviates from $1/r$, but converges with increasing r_c .
- aa : Artificial dipole correlation ($r_m > L/2$).
- $\Rightarrow$ Accurate dipole fluctuation: $zm, 1$ only.

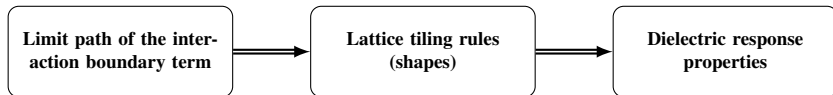
$$u_{pair}^{zm,0}(r) = \frac{1}{r} - \frac{1}{r_c} \quad (r \leq r_c)$$



$$u_{pair}^{zm,1}(r) = \frac{1}{r} + \frac{r^2}{2r_c^3} - \frac{3}{2r_c} \quad (r \leq r_c)$$

$$u_{pair}^{zm,2}(r) = \frac{1}{r} - \frac{15}{8r_c} + \frac{5r^2}{4r_c^3} - \frac{3r^4}{8r_c^5} \quad (r \leq r_c)$$

First property-form link:



I. Long-Range Correlation and Order of Divergence

Ornstein-Zernike :

$$h(\mathbf{r}_{12}) = c(\mathbf{r}_{12}) + \rho \int c(\mathbf{r}_{13})h(\mathbf{r}_{32}) d\mathbf{r}_3$$

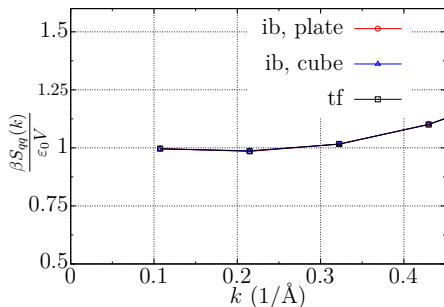
$\Downarrow$ k -space

$$S_{qq}(\mathbf{k}) = \frac{1}{V} \sum_{i=1}^N \sum_{j=1}^N q_i q_j \langle e^{i\mathbf{k} \cdot \mathbf{r}} \rangle$$

$\Downarrow$ Debye-Hückel approx

$$S_{qq}(\mathbf{k}) = \frac{k^2 I}{k^2 + \kappa_{\text{DH}}^2} = \frac{k_B T \epsilon_0}{\lambda_{\text{DH}}^2 + 1/k^2} \quad \text{at small } k$$

The correct order of divergence reflects whether a conductor can **fully screen** an external electric field (Stillinger-Lovett 2nd-moment condition⁸).



Other interactions only replacing $\mathcal{F}[1/r]$ ⁹:

$$\frac{\beta}{\epsilon_0} \frac{S_{qq}(\mathbf{k})}{k^2} = \frac{1}{\lambda_{\text{DH}}^2 k^2 + \epsilon_0 k^2 \mathcal{F}[w](k)}$$

$w(r)$ is the unit periodic extension of $u_{\text{pair}}(\mathbf{r})$

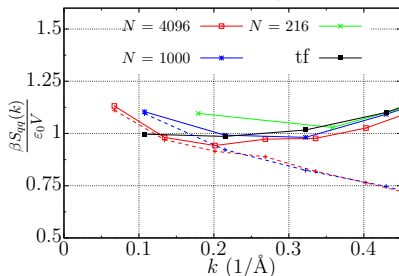
⁸F. H. Stillinger, Jr. and R. Lovett, *J. Chem. Phys.* **48**, 3858 (1968)

⁹Z. Hu, *J. Chem. Phys.* **156**, 034111 (2022)

I. Long-Range Correlation and Order of Divergence

Other interactions all differ in divergence order from $\mathcal{F}[1/r] = \frac{1}{\epsilon_0 k^2}$.

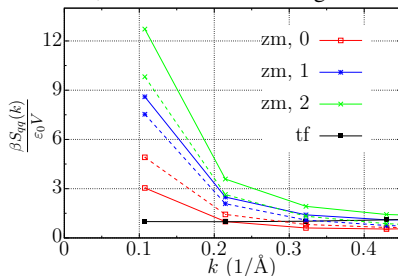
solid line: simulation values, dashed line: formula, N : number of charges



$$\lim_{k \rightarrow 0} \mathcal{F}[w_{aa}] = \frac{r_m^2}{10\epsilon_0}$$

none diverge as $1/k^2$

but both r_m and $\mathbf{k}$ vary with L , hence $\beta S_{qq}/(\epsilon_0 k^2)$ always stays near the correct value 1.



$$\lim_{k \rightarrow 0} \mathcal{F}[w_{zm,0}] = \frac{r_c^2}{6\epsilon_0}$$

$$\lim_{k \rightarrow 0} \mathcal{F}[w_{zm,1}] = \frac{r_c^2}{10\epsilon_0}$$

$$\lim_{k \rightarrow 0} \mathcal{F}[w_{zm,2}] = \frac{r_c^2}{14\epsilon_0}$$

r_c is independent of L , and diverges completely when $L \rightarrow \infty$.

I. Long-Range Correlation and Order of Divergence

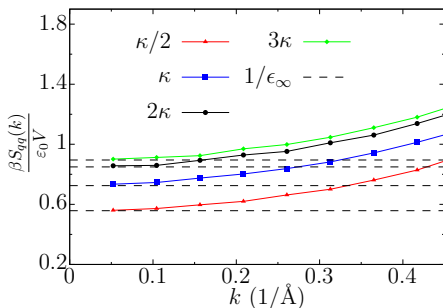
We generalize it to systems with electronic degrees of freedom:

adiabatic :

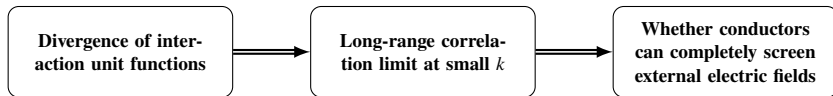
$$\frac{\beta}{\epsilon_0} \frac{S_{\text{adiabatic}}(\vec{r}, \vec{e}_0)}{k^2} = \frac{\epsilon - 1}{\epsilon} - \frac{\epsilon_\infty - 1}{\epsilon_\infty}$$

non-adiabatic :

$$\frac{\beta}{\epsilon_0} \frac{S_{\text{non-adiabatic}}(\vec{r}, \vec{e})}{k^2} = \frac{\epsilon - 1}{\epsilon}$$



Second property-form link:



I . Virial Theorem in PBC and Homogeneity

Virial Theorem:

$$P = k_B T \frac{1}{Z} \frac{\partial Z}{\partial V}, \quad Z = \int_V \cdots \int_V e^{-\beta U} d\mathbf{r}_1 \dots d\mathbf{r}_N$$

general case

$$U(\mathbf{r}_1, \dots, \mathbf{r}_N)$$

$$P = \frac{Nk_B T}{V} + \frac{1}{3V} \left\langle \sum_i^N \mathbf{r}_i \cdot \mathbf{F}_i \right\rangle$$

under PBC

$$U(\mathbf{r}_1, \dots, \mathbf{r}_N) \Rightarrow U(\{\mathbf{r}_i\}, V)$$

$$P = \frac{Nk_B T}{V} + \frac{1}{3V} \left\langle \sum_i^N \mathbf{r}_i \cdot \mathbf{F}_i \right\rangle - \left\langle \frac{\partial U(\{\mathbf{r}_i\}, V)}{\partial V} \right\rangle$$

Starting from the complete electrostatic potential $u_{\text{pair}}(\mathbf{r})$:

$$P = \frac{Nk_B T}{V} - \left\langle \frac{\partial U}{\partial V} \right\rangle = \frac{Nk_B T}{V} + \left\langle \frac{U}{3V} \right\rangle$$

Originating from the L^{-1} -homogeneity of all length quantities in u_{pair} .

$$u_{\text{pair}}(\mathbf{r}) = \frac{1}{r} + \sum_{\mathbf{n} \neq 0} \left(\frac{1}{|\mathbf{r} + \mathbf{nL}|} - \frac{1}{|\mathbf{nL}|} \right)$$

Therefore, this relation serves as the strict criterion for pressure in periodic systems.

I . Virial Theorem in PBC and Homogeneity

Ewald interaction class :

$u_{\text{pair}}^{ib,shape}(\mathbf{r})$: a rearrangement of $u_{\text{pair}}(\mathbf{r})$ satisfies

$u_{\text{pair}}^{tf}(\mathbf{r})$: a rearrangement of $u_{\text{pair}}(\mathbf{r}) - u_{\text{pair}}^{ib}(\mathbf{r})$ satisfies

aa:

$$u_{\text{pair}}^{aa} = \left(\frac{1}{r} + \frac{r^2}{2r_m^3} - \frac{3}{2r_m} \right) + \sum_{\mathbf{n} \neq 0} \left(\frac{1}{|\mathbf{r} + \mathbf{nL}|} + \frac{|\mathbf{r} + \mathbf{nL}|^2}{2r_m^3} - \frac{3}{2r_m} \right) \quad \text{satisfies}$$

zm, 1:

$$u_{\text{pair}}^{zm,1} = \left(\frac{1}{r} + \frac{r^2}{2r_c^3} - \frac{3}{2r_c} \right) + \sum_{\mathbf{n} \neq 0} \left(\frac{1}{|\mathbf{r} + \mathbf{nL}|} + \frac{|\mathbf{r} + \mathbf{nL}|^2}{2r_c^3} - \frac{3}{2r_c} \right), \quad \text{violates}$$

difference: $r_m \propto L^{-1}$, r_c fixed

Thus, any r -space cutoff must $\propto L^{-1}$;

otherwise, pressure artifacts arise, as seen in Zero-Multipole¹⁰.

¹⁰H. Wang, X. Gao, and J. Fang, *J. Chem. Theory Comput.* **12**, 5596–5608 (2016). 

I . Virial Theorem in PBC and Homogeneity

Pressure correction in one-component plasma :

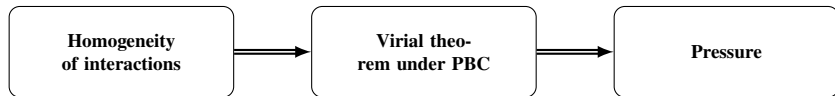
$$U^{corr} = -\frac{\pi \left(\sum_{i=1}^N q_i \right)^2}{2\alpha^2 V}, \quad P^{corr} = -\left\langle \frac{\partial U^{corr}}{\partial V} \right\rangle$$

originates from the neutralizing background (the **Jellium**) via $\text{erfc}(\alpha r)/r$.

$$U^{corr} \text{ is a part of } \frac{1}{2} \sum_{i < j} q_i q_j u_{\text{pair}}(\mathbf{r}_{ij}) - E_{\text{traditional Ewald}}$$

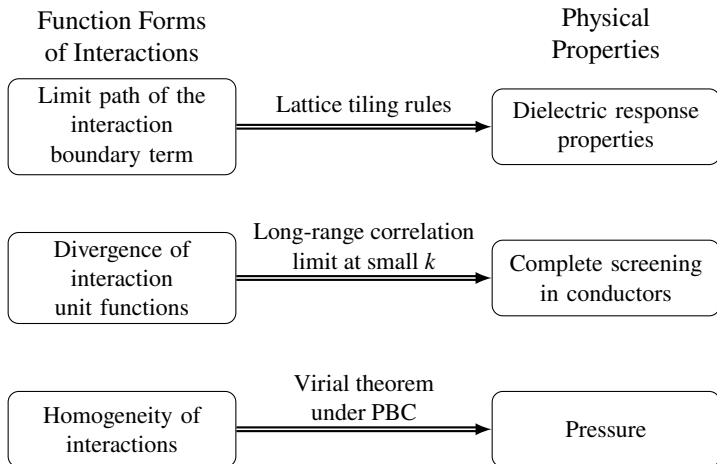
The neutralizing background is a non-physical entity, a byproduct of the Ewald construction.

Third property-form link:



I . Summary

A Unified Theoretical Framework



II. A PME without Systematic Error

Building upon Ewald's concept and focusing on computational efficiency has led to:

- ▶ A family of particle-mesh methods (PME, SPME, PPPM).
- ▶ Yet, theoretical inaccuracies and technical issues (e.g., global FFT communication) limit larger-scale simulations (e.g., D. E. Shaw's Anton).

Thus, analyzing and eliminating these errors is essential.

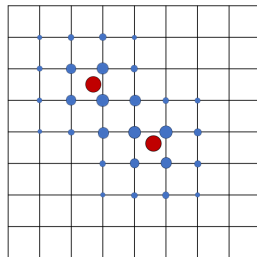
II. Particle Mesh Ewald

PME ($\mathcal{O}(N \lg N)$) is a fast algorithm for Ewald summation ($\mathcal{O}(N^2)$)^{11,12}

$$E^k = \frac{1}{2V} \sum_{\mathbf{k} \neq 0} \frac{4\pi}{k^2} e^{-\frac{k^2}{4\alpha^2}} \left(\sum_{i=1}^N q_i e^{i\mathbf{k}\mathbf{r}_i} \right) \left(\sum_{j=1}^N q_j e^{-i\mathbf{k}\mathbf{r}_j} \right)$$

Interpolating $e^{ikx} = \text{charge interpolation} + \text{FFT}$:

$$\begin{aligned} q_j e^{ik_1 x_j} &\approx \sum_{n_1=0}^{K_1-1} q_j W_p(x_j - n_1 \frac{L_x}{K_1}) e^{ik_1 n_1 \frac{L_x}{K_1}} \\ &= \text{FFT1D}[q_j W_p(x_j - n_1 \frac{L_x}{K_1})](m_1) \end{aligned}$$



W_p : interpolation coefficient of order p .

$$E_{\text{PME}}^k = \frac{1}{2} \sum_{m_1=0}^{K_1-1} \sum_{m_2=0}^{K_2-1} \sum_{m_3=0}^{K_3-1} \frac{1}{V} \frac{4\pi}{k^2} e^{-\frac{k^2}{4\alpha^2}} \text{FFT3D}[Q](m_1, m_2, m_3) \cdot \text{FFT3D}[Q](-m_1, -m_2, -m_3)$$

¹¹T. Darden and L. Pedersen, J. Chem. Phys. **98**, 10089(1993)

¹²U. Essmann and L.G. Pedersen, J. Chem. Phys. **103**, 8577(1995)

II. Truncation Error and It's Elimination

Wavevector cutoff in interactions couples to mesh size

$K_1 \times K_2 \times K_3$:

$$u^k(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{k} \in \mathbf{k}_{\text{cut}}} \frac{4\pi}{k^2} e^{-\frac{k^2}{4\alpha^2}} e^{i\mathbf{k} \cdot \mathbf{r}}$$

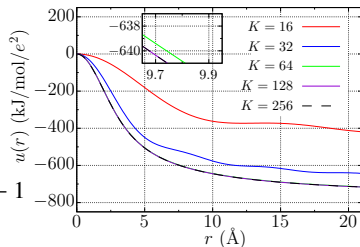
Wavevector :

$$\mathbf{k}(m_1 \frac{2\pi}{L_x}, m_2 \frac{2\pi}{L_y}, m_3 \frac{2\pi}{L_z}), \quad \frac{K}{2} \geq m \leq \frac{K}{2} - 1$$

So:

$u^k(\mathbf{r})$ (small $K_1 \times K_2 \times K_3$) differs from $u^k(\mathbf{r})$ (large $K_1 \times K_2 \times K_3$)

This is the truncation error in PME.



Truncation error is the main reason that restricts mesh size to satisfy

$$L/K \approx 1 \text{ \AA} .$$

II. Truncation Error and Its Elimination

Decoupling wavevector cutoff range from mesh size:

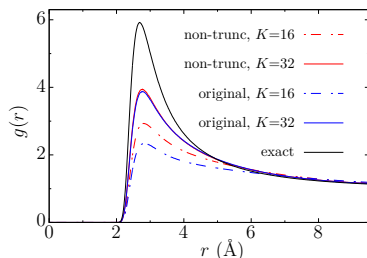
$$u_{K'}^k(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{k} \in \mathbf{K}'} \frac{4\pi}{k^2} e^{-\frac{k^2}{4\alpha^2}} e^{i\mathbf{k} \cdot \mathbf{r}}$$

K' is enough to make $u_{K'}^k$ converge. Corresponding non-trunc PME:

$$E_k^{\text{PME}} = \frac{1}{2} \sum_{m_1=0}^{K_1-1} \sum_{m_2=0}^{K_2-1} \sum_{m_3=0}^{K_3-1} \left(\text{FFT3D} \left[\frac{1}{K'_1 K'_2 K'_3} u_{K'} \right] \text{FFT3D}[Q](\mathbf{m}) \cdot \text{FFT3D}[Q](-\mathbf{m}) \right)$$

charge exactly on grid

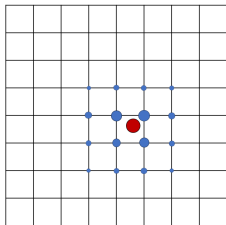
$K_1 \cdot K_2 \cdot K_3$	$E_k^{\text{PME}} (\text{kJ/mol})$	
	ori-PME	non-PME
16 · 16 · 16	441.5625	745.7720
32 · 32 · 32	675.6669	745.7720
64 · 64 · 64	744.8545	745.7720
128 · 128 · 128	745.7720	745.7720



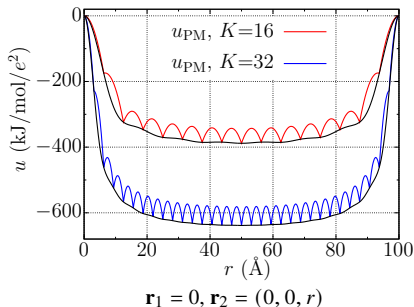
simulation sampling

II. Artifact Error and It's Elimination

Interpolating each single charge produces artifacts, where part of the interactions are redundant (but not all).



Blue points are artifacts



Defining PME interactions in two-charge systems :

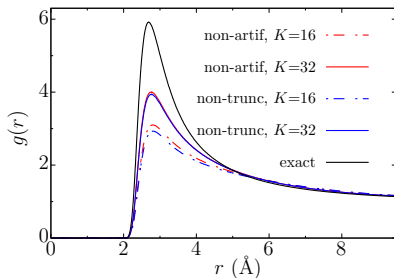
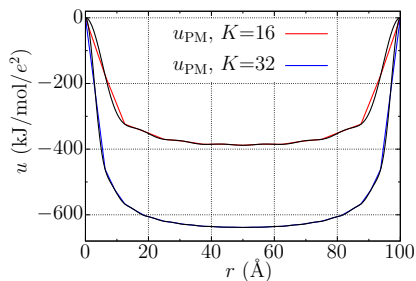
$$u_{\text{PM}}^k(\mathbf{r}_1, \mathbf{r}_2) = \frac{E_{\text{non-trunc}}^k(\mathbf{r}_1, \mathbf{r}_2)}{q_1 q_2}$$

This is the artifact error in PME.

Artifact error cause regular oscillations in PME interactions.

II. Artifact Error and It's Elimination

Directly removing the redundant interaction energy yields non-artifact PME, but the difficulty lies in that each single point charge's artifact system is non-electroneutral.



Calculating E_i via non-neutral Ewald/PME :

$$E_{\text{non-artif}}^k = E_{\text{non-trunc}}^k - \sum_{i=1}^N E_i^k (q_i \text{ coordinates relative to the grid})$$

Corresponding to non-artif PME.

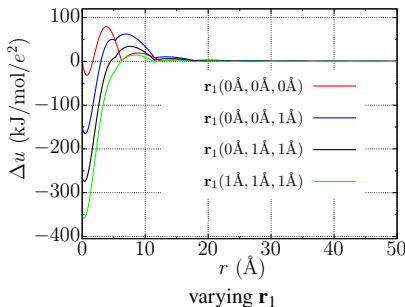
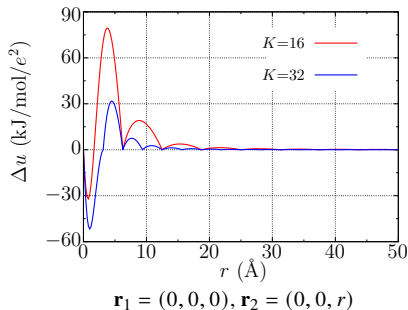
Higher-order interpolation follows the same principle as linear but is more complex.

II. Fitting Error of Interactions and It's Elimination

Perspective shift: from discrete charge interactions to original charge interactions

$$u_{\text{PM}}^{\text{non-artif}}(\mathbf{r}_1, \mathbf{r}_2) = \sum_{\mathbf{m}} W_p(\mathbf{r}_1, \mathbf{m}) \sum_{\mathbf{n}} W_p(\mathbf{r}_2, \mathbf{n}) \cdot u_{\text{pair}}(\mathbf{r}_{\text{grid}}(\mathbf{m}) - \mathbf{r}_{\text{grid}}(\mathbf{n}))$$

which deviates from exact results and depends independently on $\mathbf{r}_1$ and $\mathbf{r}_2$.



This is the fitting error of interactions.

Meanwhile, the r -space interaction of the original charge remains unchanged.

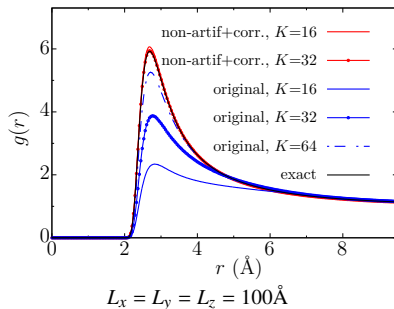
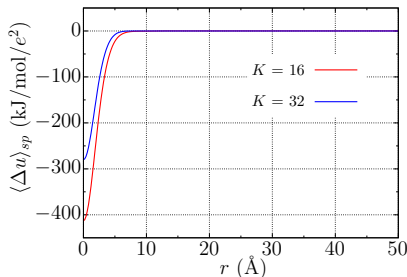
Fitting error causes Ewald's splitting of $1/r$ to be no longer complete.

II. Fitting Error of Interactions and It's Elimination

Eliminate all unphysical dependencies through "translation-rotation averaging":

$$\langle \Delta u \rangle_{sp} == \frac{1}{2\pi^2\Omega} \int_{\Omega} d\mathbf{r}_1 \int_0^{\pi} d\theta \int_0^{2\pi} d\varphi [u_{\text{PM}}^{\text{non-artif}} - u_{\text{pair}}](\mathbf{r}_1, \theta, \varphi)$$

Merge $\langle \Delta u \rangle_{sp}$ into the original r -space interaction to ensure complete splitting.



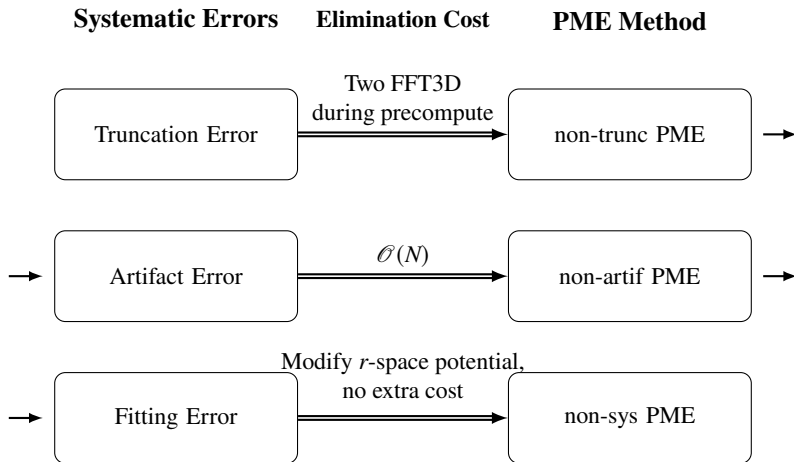
non-sys PME: accurate at $16 \times 16 \times 16$ (original PME requires $128 \times 128 \times 128$)

Remaining **unbiased noise** requires $\mathcal{O}(N^2)$, conflicting with $\mathcal{O}(N \log N)$.

Thus, we consider the unbiased noise as the **minimal cost** of PME calculation.

II. Summary

A PME Without Systematic Errors



Non-sys PME: 100× faster (90× theoretical) via mesh sparsification, with significantly reduced memory.

Furthermore, we argue that the conflict between **momentum** and **energy** conservation in PME is irreconcilable.

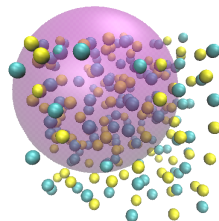
III. Toward MLFFs without Systematic Error

Map **local** environment to energy/forces

$$E_{\text{PBC}} = \sum_{i=1}^N E_{\text{vacuum}}(\mathbf{r}_i; r_c)$$

This inherently contradicts long-range electrostatics.

Therefore, long- and short-range interactions must be separated in ML training data.



With rigorous separation,
ML is already sufficiently powerful
(Accuracy: 10^{-1} meV/Å).

ML	Drude Model	
	$\langle F_i^{\text{ML}} - F_i \rangle (\text{meV}/\text{\AA})$	
	Na-core	Cl-core
exact-BPNN	1.713×10^{-7}	1.713×10^{-7}
DPMD-short	3.695×10^{-1}	4.238×10^{-1}
DPMD-full	22.086	18.946

First two lines: short-range forces only

Last line: total force

Thus, correct separation of long-range and short-range interactions is key.

III. Toward MLFFs without Systematic Error

Current separation fails because:

- ▶ *ab initio* forces originate from delocalized electron clouds, not atom-centered point charges
- ▶ Separation replaces true density with point charges (nuclei + Wannier centers)

Therefore, rigorous separation must be based on precise atom-assignable electron density distribution.

Based on the same methodology as X-ray spectrum analysis:

- ▶ Solve bulk total X-ray spectrum and static structure factor
- ▶ Obtain atom-assignable density via inverse transform with physically constrained form factors

**This enables rigorous long- and short-range separation in training data
— a crucial step toward MLFFs without Systematic Error.**

III. Toward MLFFs without Systematic Error

In adiabatic dynamics, even without long-range interactions, systematic errors persist:

- ▶ Electronic self-consistency is intrinsically global. Even in purely short-range systems, local equilibrium $\neq$ global equilibrium
- ▶ In essence, this indirect correlation is also long-ranged

Implication: Different local environments should exchange information when mapping to atomic forces — analogous to FFT's global communication.

Therefore, the nonlocality of both long-range and indirect correlations is the primary source of systematic errors in MLFFs.